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Multiple-barrel firearm and method of use

Abstract

A machine gun comprising a gun housing; a rotor assembly positioned within the gun housing; a barrel cluster including a plurality of barrels extending forward from the rotor assembly; one or more vents providing a path for expulsion away from the rotor assembly of gases emanating from a cartridge chambered in one of the plurality of barrels; and one or more of a rotor lock mechanism, a feed mechanism, a delinking and transfer mechanism, a searing mechanism, a drive system, a self-contained hardware barrel clamp, a threaded adapter barrel clamp, integrated mounts, and/or multiple feed inlet locations.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates in general to multiple-barrel firearms and in particular to electric motor-driven multiple-barrel firearms.

SUMMARY OF THE INVENTION

(2) An aspect of the invention involves a multiple-barrel firearm that has similar barrel spacing, bolt assemblies, and bolt cam path as existing M134 platforms, but utilizes alternative ammunition feeding, delinking, and transfer mechanisms. The multiple-barrel firearm is nearly half the weight of any existing M134 platform and utilizes a mechanical barrel cluster rotational lock for absolute safety when the system is in safe condition. The drive system utilizes a brushless DC servo motor

with closed loop positional feedback for precise control of all rotating components. The multiple-barrel firearm has a longitudinal bolt searing safety mechanism, self-contained hardware barrel clamp with threaded adapter for various muzzle devices, integrated suspension lug mounting provisions, as well as multiple feed inlet locations for optimal feed chute orientation.

(3) Another aspect of the invention involves a machine gun comprising a gun housing; a rotor assembly positioned within the gun housing; a barrel cluster including a plurality of barrels extending forward from the rotor assembly; one or more vents providing a path for expulsion away from the rotor assembly of gases emanating from a cartridge chambered in one of the plurality of barrels; and one or more of a rotor lock mechanism, a feed mechanism, a delinking and transfer mechanism, a searing mechanism, a drive system, a self-contained hardware barrel clamp, a threaded adapter barrel clamp, integrated mounts, and/or multiple feed inlet locations.

(4) One or more implementations of the aspect of the invention described immediately above includes one or more of the following: a motor configured to rotate the rotor assembly, a motor housing, and the rotor lock mechanism configured to lock the rotor assembly relative to the motor housing, preventing any rotation of the barrel cluster to ensure safety; a bonded clutch solenoid, an external control interface, and a gun control module including a processor and one or more software modules that are configured to, when executed by the processor, respond to external control signals to disengage bonded clutch solenoid, stopping rounds from being fed into the rotor assembly, the gun control module further configured to control the motor to run any remaining rounds through the firing cycle and spin the barrel cluster multiple times to ensure chambers are clear, and electrically safe the machine gun; the feed mechanism includes a feeder sprocket that is concentric to and rotates in an opposite direction of the barrel cluster so as to cause an inertial load of the feed mechanism to counteract an inertial load of the barrel cluster; the delinking and transfer mechanism including a reciprocating delinking shuttle and a barrel cam operably associated with the reciprocating delinking shuttle to control linear movement of a reciprocating delinking shuttle to delink and transfer each round; the searing mechanism having a longitudinally movable bolt searing ramp, allowing additional safety to control firing pin action; the drive system having a feedback control loop with a brushless DC servo motor, a position encoder sensor, and a programmable gun control module, allowing variable rate of fire, and current and temperature sensing to offer instantaneous system status and motor control; the motor configured to rotate the rotor assembly and a motor housing with cooling features, transferring excess heat to ambient environment; the self-contained hardware barrel clamp configured to prevent each barrel from rotating about its own axis; a lock housing with a slot, a lock plunger, and the spring-loaded rotating barrel lock whereby the barrel clamp is removed via tool inserted into the slot so that force is applied to the lock plunger, rotating spring-loaded rotating barrel lock, and allowing the barrel clamp to slide off the barrel cluster; the plurality of barrels include respective non-threaded barrel end portions and the machine gun further including the threaded adapter barrel clamp with threads to mount a multitude of muzzle devices to the non-threaded barrel end portions; one or more stabilizer adapters; the one or more stabilizer adapters include integrated mounts for suspension lugs; the one or more stabilizer adapters include load transferring pads; the feed mechanism includes a feed mechanism housing configured to allow for multiple feed inlet locations to optimize feed chute orientation based on system application; and/or the feed mechanism includes a feed chute inlet, a feeder door adapter, and one or more feeder closeouts, the feed mechanism housing includes a bottom, sides, and a top, and the feed mechanism housing is configured to allow for the feed chute inlet at the bottom of the feed mechanism housing with the feeder door adapter mounted to a lowest position and the one or more feeder closeouts located in the top of the feed mechanism housing, allow for the feed chute inlet at one of the sides of the feed mechanism housing with feeder door adapter mounted to one of the sides of the feed mechanism housing and the one or more feeder closeouts located in remaining open slots of the feed mechanism housing, and allow for the feed chute inlet at the top of the feed mechanism housing with feeder door adapter mounted

to the top of the feed mechanism housing and the one or more feeder closeouts located in remaining open slots of the feed mechanism housing.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The accompanying drawings, which are incorporated in and form a part of this specification illustrate embodiments of the invention and together with the description, explain the principles of the invention.
- (2) FIG. 1 is a perspective view of an embodiment of a multiple-barrel firearm.
- (3) FIG. 2 is an exploded perspective view of the multiple-barrel firearm of FIG. 1.
- (4) FIG. 3 is a front view of a prior art M134G multiple-barrel firearm.
- (5) FIG. 4 is a rear view of the prior art M134G multiple-barrel firearm of FIG. 3.
- (6) FIG. 5 illustrates cross-sectional views of a rotor lock mechanism of the multiple-barrel firearm of FIG. 1.
- (7) FIG. 6 is a cross-sectional view of a feed mechanism of the multiple-barrel firearm of FIG. 1.
- (8) FIG. 7 is a perspective view of a delinking and transfer mechanism of the multiple-barrel firearm of FIG. 1.
- (9) FIG. 8 illustrates perspective views of a searing mechanism of the multiple-barrel firearm of FIG. 1.
- (10) FIG. 9 illustrates perspective/cross-sectional views of a drive system of the multiple-barrel firearm of FIG. 1.
- (11) FIG. 10 is a cross-sectional view of a self-contained hardware barrel clamp of the multiple-barrel firearm of FIG. 1.
- (12) FIG. 11 is a perspective view of a threaded adapter barrel clamp of the multiple-barrel firearm of FIG. 1.
- (13) FIG. 12 is a cross-sectional view of integrated mounts for suspension lugs of the multiple-barrel firearm of FIG. 1.
- (14) FIG. 13 illustrates perspective views of multiple feed inlet locations of the multiple-barrel firearm of FIG. 1.
- (15) FIG. 14 is a cross-sectional view of gun housing of the multiple-barrel firearm of FIG. 1.
- (16) FIG. 15 is an electrical block diagram of an embodiment of electronic components of the multiple-barrel firearm of FIG. 1.

DESCRIPTION OF EMBODIMENT OF THE INVENTION

- (17) With reference to FIGS. 1-15, an embodiment of a multiple-barrel firearm/machine gun **1000** will be described. The machine gun **1000** includes a gun housing **1174**; a rotor assembly **1030** positioned within the gun housing **1174**; a barrel cluster **1004** including a plurality of barrels **1350** extending forward from the rotor assembly **1030**; one or more vents **1176** providing a path for expulsion away from the rotor assembly **1030** of gases emanating from a cartridge chambered in one of the plurality of barrels **1350**. The machine gun **1000** includes a rotor lock mechanism **1002** (FIGS. 2, 5), a feed mechanism **1048** (FIGS. 2, 6), a delinking and transfer mechanism **1100** (FIGS. 2, 7), a searing mechanism **1192** (FIGS. 2, 8), a drive system **1252** (FIGS. 2, 9, 15), a self-contained hardware barrel clamp **1360** (FIGS. 2, 10), a threaded adapter barrel clamp **1362** (FIGS. 2, 11), integrated mounts **1382** (FIGS. 2, 12), multiple feed inlet locations **1422** (FIGS. 2, 13), each of which will be described in turn below.
- (18) With reference to FIG. 5, a rotor lock mechanism **1002** of the multiple-barrel machine gun **1000** will be described. The machine gun **1000** includes a motor (e.g., brushless DC servo motor) **1254** configured to rotate the rotor assembly **1030**, a motor housing **1256** (with aft motor housing **1022**), and the rotor lock mechanism **1002**, which is configured to lock the rotor assembly **1030**

relative to the motor housing **1256**, preventing any rotation of the barrel cluster **1004** to ensure safety. The rotor lock mechanism **1002** includes a mechanical rotational lock **1006** that activates when the operator sets the machine gun **1000** to SAFE to ensure machine gun **1000** is in an absolute safe condition. The rotor lock mechanism **1002** includes a barrel cluster rotational lock mount **1020**, rotor lock plunger **1024**, spring **1032**, and rotor assembly holes **1040**. The barrel cluster rotational lock mount **1020** is fixed to the aft motor housing **1022**, which precisely locates rotor lock shuttle **1010** to allow for translational movement of rotor lock plunger **1024** to be either engaged or disengaged with rotor assembly **1030**. In SAFE condition, the rotor lock plunger **1024** is spring-loaded via spring **1032** to engage holes **1040** in rotor assembly **1030**, preventing any rotation of barrel cluster to ensure absolute safety. When the machine gun **1000** is set to ARMED condition, the rotor lock shuttle **1010** is spring-loaded open via the spring **1032**, ensuring the rotor lock plunger **1024** is clear of rotating rotor assembly **1030**, allowing for rotation of the barrel cluster **1004** to fire rounds. The rotor lock mechanism **1002** locks the rotor assembly **1030** relative to the motor housing **1256**.

(19) With reference to FIG. 6, a feed mechanism **1048** of the multiple-barrel machine gun **1000** will be described. The feed mechanism **1048** includes a feeder sprocket **1080** that is concentric to and rotates in an opposite direction of the barrel cluster **1004** so as to cause an inertial load of the feed mechanism **1048** to counteract an inertial load of the barrel cluster **1004**, resulting in nearly zero rotational inertial load at mounting location **1049** (FIG. 2) as the inertial loads of feed mechanism **1048** and barrel cluster **1004** counteract each other. The feed mechanism **1048** includes ammo retaining lever **1060**, feeder door **1070**, feed chute adapter inlet **1050**, feeder closeout panels **1090**, and exhaust chute **1110**. With ammo retaining lever **1060** and feeder door **1070** in an open position, rounds are loaded into the firearm **100** system at feed chute adapter inlet **1050**. The ammo retaining lever **1060** is then closed, holding rounds inside feeder sprocket **1080** until the feeder door **1070** is closed. Rounds will rotate inside feeder sprocket **1080** while being guided by feeder closeout panels **1090** around machine gun **1000** until they reach delinking and transfer mechanism **1100**. After ammunition has been delinked, the rounds will be transferred into rotor assembly **1030** and the remaining links continue rotating inside feeder sprocket **1080** until they are disposed of through the link exhaust chute **1110**.

(20) With reference to FIG. 7, the delinking and transfer mechanism **1100** of the multiple-barrel machine gun **1000** will be described. The delinking and transfer mechanism **1100** includes a reciprocating delinking shuttle **1170** and a barrel cam **1180** operably associated with the reciprocating delinking shuttle **1170** to control linear movement of a reciprocating delinking shuttle **1170** to delink and transfer each round into the bolt face **1172** (FIG. 14). The delinking and transfer mechanism **1100** includes meshing gears **1150**, transfer sprocket **1140**, guide bar **1130**, precision locating boss **1190**, and spring-loaded quick-release over-center locking latch **1160**. The delinking and transfer mechanism **1100** is driven off the barrel cluster rotor assembly **1030** using the meshing gears **1150**. The gears **1150** continuously rotate barrel cam **1180** which controls delinking shuttle **1170** in a linear reciprocating motion to transfer a delinked round from the feed mechanism **1048** into transfer sprocket **1140**. The round is then rotated around the transfer sprocket **1140** and controlled by guide bar **1130** to increase tangential velocity for handoff into the bolt face **1172**. The delinking and transfer mechanism **1100** is attached to the gun housing **1174** by precision locating boss **1190** and held into place using the spring-loaded quick-release over-center locking latch **1160**.

(21) With reference to FIG. 8, a searing mechanism **1192** of the multiple-barrel machine gun **1000** will be described. The searing mechanism **1192** includes a longitudinally movable bolt searing ramp **1210**, allowing additional safety to control firing pin action. The searing mechanism **1192** includes initial ramp **1200**, spring-loaded knob **1250**, ramp **1210**, flat path **1220**, safe position **1240**, and bolt ramp slide **1230**. The searing mechanism **1192** utilizes longitudinal translation of the bolt searing ramp **1210** to allow operator additional safety to control the firing pin action. After a delinked round is transferred into the bolt face **1172**, the bolt follows the initial ramp **1200** to move

round into barrel chamber **1202** (FIG. **14**). If spring-loaded knob **1250** is in the fire position as shown in FIG. **8**, the bolt will further follow searing ramp **1210** to sear the bolt, firing the round. The fired round will dwell in the barrel chamber **1202** as bolt follows flat path **1220** prior to extraction. If spring-loaded knob **1250** is in the SAFE position **1240**, bolt ramp slide **1230** will move rearwards, causing the ramp **1210** to no longer sear the bolt as max position will coincide with end of initial ramp **1200**.

(22) With reference to FIGS. **9** and **15**, a drive system **1252** of the multiple-barrel machine gun **1000** will be described. The drive system **1252** includes a feedback control loop with brushless DC servo motor **1254** to rotate the rotor assembly **1030**, position encoder sensor **1290**, and a programmable drive, allowing variable rate of fire, and current and temperature sensing to offer instantaneous system status and motor control. Motor housing **1256** includes cooling features **1300**, transferring excess heat to ambient environment to keep the motor temperature within operating range. The motor **1254** includes a rotating shaft **1260** with a keyed spline to ensure proper round handoff timing. The motor **1254** includes a stator bonded to motor housing **1022** and includes provisions for a bonded clutch solenoid **1280**. The position encoder sensor **1290** is attached to the motor's rotating shaft **1260** and the motor housing **1022**. The drive system **1252** includes an external control interface **1302**, and a gun control module/processor **1018** including one or more software modules that are configured to, when executed by the gun control module/processor **1018**, respond to external control signals to FIRE and electrically SAFE the weapon, engaging and disengaging the bonded clutch solenoid **1280**, stopping rounds from being fed into rotor assembly. The gun control module **1018** controls the motor **1254** to run any remaining rounds through the firing cycle and spin the barrel cluster **1004** multiple times to ensure chambers **1202** are clear.

(23) With reference to FIG. **10**, a self-contained hardware barrel clamp **1360** of the multiple-barrel machine gun **1000** will be described. The self-contained hardware barrel clamp **1360** has no removable hardware, can be removed in a single motion with a screwdriver or similar tool, and is configured to prevent barrel cluster **1004** as a whole from rotating about its own axis. The barrel clamp **1360** includes a lock housing **1330** with a slot **1310**, a lock plunger **1320**, and a spring-loaded rotating barrel lock **1340** whereby the barrel clamp **1360** is removed via a tool (e.g., screwdriver) inserted into the slot **1310** so that force is applied to the lock plunger **1320**, rotating spring-loaded rotating barrel lock **1340**, and allowing the barrel clamp **1360** to slide off the barrel cluster **1004**. The barrel clamp **1360** is located against feature **1362** on barrel **1350**. The barrel clamp lock housing **1330** contains provisions for barrel locks **1340** and pre-loaded compression springs **1342** to prevent barrel clamp **1360** from sliding forward on barrels **1350**. The barrel locks **1340** have an offset center of gravity from pivot **1364** to utilize centrifugal force to ensure barrel clamp **1360** is locked to barrels **1350**. During removal of the barrel clamp **1360**, the screwdriver or similar tool is inserted into one of the slots **1310** of lock housing **1330**, then force is applied to lock plunger **1320**, rotating barrel locks **1340** while also compressing springs **1342**, unlocking the barrel clamp **1360**, and allowing it to slide off the barrel cluster **1004**.

(24) With reference to FIG. **11**, a threaded adapter barrel clamp **1362** of the multiple-barrel machine gun **1000** will be described. The threaded adapter barrel clamp **1360** includes standard threaded features/threads **1380** to mount a multitude of muzzle devices **1370** (e.g., flash hiders, sound suppressors, other threaded muzzle devices) to non-threaded barrel end portions **1390** of the plurality of barrels **1350**.

(25) With reference to FIG. **12**, integrated mounts **1382** for suspension lugs **1410** of the multiple-barrel machine gun **1000** will be described. The gun housing **1174** includes one or more stabilizer adapters **1400**, **1420** with the integrated mount(s) **1382** for direct integration of the suspension lug(s) **1410** for fixed forward applications. When mounted to a munitions release unit, forward stabilizer adapter **1400** includes pads **1412** to transfer any loads between release unit **1414** and main gun housing **1174**. Aft stabilizer adapter **1420** also includes transfer pads **1412**, while also containing the weapon's arm/safing sector mechanism. Most other gatling guns require

intermediate hardback components to interface between weapon system and suspension lugs, adding unnecessary weight to the overall weapon system.

(26) With reference to FIG. 13, the feed mechanism 1048 includes a feed mechanism housing 1452 configured to allow for multiple feed inlet locations 1422 to optimize feed chute orientation based on system application. The feed mechanism 1048 includes a feed chute inlet 1440, a feeder door adapter 1450, and one or more feeder closeouts 1430. For the majority of crew serve applications, the feed mechanism housing 1452 is configured to allow for the feed chute inlet 1440 at a bottom of the feed mechanism housing 1452 with feeder door adapter 1450 mounted to a lowest position and the one or more feeder closeouts 1430 located in a top of the feed mechanism housing 1452. For fixed forward applications as well as some crew serve applications, the feed mechanism housing 1452 is configured to allow for the feed chute inlet 1440 at one of the sides of the feed mechanism housing 1452 with feeder door adapter 1450 mounted to one of the sides of the feed mechanism housing 1452 and one or more feeder closeouts 1430 located in remaining open slots of the feed mechanism housing 1452, or the feed mechanism housing 1452 is configured to allow for the feed chute inlet 1440 at the top of the feed mechanism housing 1452 with feeder door adapter 1450 mounted to the top of the feed mechanism housing 1452 and the one or more feeder closeouts 1430 located in remaining open slots of the feed mechanism housing 1452.

(27) The above description of the disclosed embodiments is provided to enable any person skilled in the art to make or use the invention. Various modifications to these embodiments will be readily apparent to those skilled in the art, and the general principles described herein can be applied to other embodiments without departing from the spirit or scope of the invention. Thus, it is to be understood that the description and drawings presented herein represent a presently preferred embodiment of the invention and are therefore representative of the subject matter which is broadly contemplated by the present invention. It is further understood that the scope of the present invention fully encompasses other embodiments that may become obvious to those skilled in the art and that the scope of the present invention is accordingly not limited.

(28) The figures may depict exemplary configurations for the invention, which is done to aid in understanding the features and functionality that can be included in the invention. The invention is not restricted to the illustrated architectures or configurations, but can be implemented using a variety of alternative architectures and configurations. Additionally, although the invention is described above in terms of various exemplary embodiments and implementations, it should be understood that the various features and functionality described in one or more of the individual embodiments with which they are described, but instead can be applied, alone or in some combination, to one or more of the other embodiments of the invention, whether or not such embodiments are described and whether or not such features are presented as being a part of a described embodiment. Thus the breadth and scope of the present invention, especially in the following claims, should not be limited by any of the above-described exemplary embodiments.

(29) Terms and phrases used in this document, and variations thereof, unless otherwise expressly stated, should be construed as open ended as opposed to limiting. As examples of the foregoing: the term “including” should be read as mean “including, without limitation” or the like; the term “example” is used to provide exemplary instances of the item in discussion, not an exhaustive or limiting list thereof; and adjectives such as “conventional,” “traditional,” “standard,” “known” and terms of similar meaning should not be construed as limiting the item described to a given time period or to an item available as of a given time, but instead should be read to encompass conventional, traditional, normal, or standard technologies that may be available or known now or at any time in the future. Likewise, a group of items linked with the conjunction “and” should not be read as requiring that each and every one of those items be present in the grouping, but rather should be read as “and/or” unless expressly stated otherwise. Similarly, a group of items linked with the conjunction “or” should not be read as requiring mutual exclusivity among that group, but

rather should also be read as “and/or” unless expressly stated otherwise. Furthermore, although item, elements or components of the disclosure may be described or claimed in the singular, the plural is contemplated to be within the scope thereof unless limitation to the singular is explicitly stated. The presence of broadening words and phrases such as “one or more,” “at least,” “but not limited to” or other like phrases in some instances shall not be read to mean that the narrower case is intended or required in instances where such broadening phrases may be absent.

Claims

1. A machine gun comprising: a gun housing; a rotor assembly positioned within the gun housing; a barrel cluster including a plurality of barrels extending forward from the rotor assembly; one or more vents providing a path for expulsion away from the rotor assembly of gases emanating from a cartridge chambered in one of the plurality of barrels; a motor configured to rotate the rotor assembly, a motor housing, and a rotor lock mechanism configured to lock the rotor assembly directly to the motor housing, preventing rotation of the barrel cluster.
2. A machine gun comprising: a gun housing; a rotor assembly positioned within the gun housing; a barrel cluster including a plurality of barrels extending forward from the rotor assembly; one or more vents providing a path for expulsion away from the rotor assembly of gases emanating from a cartridge chambered in one of the plurality of barrels; a feed mechanism with a feeder sprocket that is concentric to and rotates in an opposite direction of the barrel cluster so as to cause an inertial load of the feed mechanism to counteract an inertial load of the barrel cluster.
3. The machine gun of claim 2, further including a motor configured to rotate the rotor assembly, a bonded clutch solenoid, an external control interface, and a gun control module including a processor and one or more software modules that are configured to, when executed by the processor, respond to external control signals to disengage the bonded clutch solenoid, stopping rounds from being fed into the rotor assembly, the gun control module further configured to control the motor to run any remaining rounds through the firing cycle and spin the barrel cluster multiple times to ensure chambers are clear, and electrically safe the machine gun.
4. The machine gun of claim 2, further including a searing mechanism having a longitudinally movable bolt searing ramp.
5. The machine gun of claim 2, further including a drive system having a feedback control loop with a brushless DC servo motor, a position encoder sensor, and a programmable drive, allowing variable rate of fire, and current and temperature sensing.
6. The machine gun of claim 2, wherein the motor is configured to rotate the rotor assembly and a motor housing with cooling features, transferring excess heat to ambient environment.
7. The machine gun of claim 2, wherein the plurality of barrels include respective non-threaded barrel end portions and further including a threaded adapter barrel clamp with threads to mount a multitude of muzzle devices to the non-threaded barrel end portions.
8. The machine gun of claim 2, further including one or more stabilizer adapters.
9. The machine gun of claim 8, wherein the one or more stabilizer adapters include integrated mounts for suspension lugs.
10. The machine gun of claim 8, wherein the one or more stabilizer adapters include load transferring pads.
11. A machine gun comprising: a gun housing; a rotor assembly positioned within the gun housing; a barrel cluster including a plurality of barrels extending forward from the rotor assembly; one or more vents providing a path for expulsion away from the rotor assembly of gases emanating from a cartridge chambered in one of the plurality of barrels; a delinking and transfer mechanism including a non-rotating reciprocating delinking shuttle and a rotating barrel cam operably associated with the non-rotating reciprocating delinking shuttle to control linear movement of the non-rotating reciprocating delinking shuttle to delink and transfer each round.

12. A machine gun comprising: a gun housing; a rotor assembly positioned within the gun housing; a barrel cluster including a plurality of barrels extending forward from the rotor assembly; one or more vents providing a path for expulsion away from the rotor assembly of gases emanating from a cartridge chambered in one of the plurality of barrels; a self-contained hardware barrel clamp configured to prevent each barrel of the plurality of barrels from rotating about an axis thereof, wherein the barrel clamp includes a lock housing with a slot, a lock plunger, and a spring-loaded rotating barrel lock whereby the barrel clamp is removable from the barrel cluster by inserting a tool into the slot of the lock housing, applying a force to the lock plunger, rotating the spring-loaded rotating barrel lock, and sliding the barrel clamp off the barrel cluster.

13. A machine gun comprising: a gun housing; a rotor assembly positioned within the gun housing; a barrel cluster including a plurality of barrels extending forward from the rotor assembly; one or more vents providing a path for expulsion away from the rotor assembly of gases emanating from a cartridge chambered in one of the plurality of barrels; further including a feed mechanism with a feed mechanism housing configured to allow for a single feed inlet location of multiple possible feed inlet locations.

14. The machine gun of claim 13, wherein the feed mechanism includes a feed chute inlet, a feeder door adapter, and one or more feeder closeouts, the feed mechanism housing includes a bottom, sides, and a top, and the feed mechanism housing is configured to allow for the feed chute inlet at one of the bottom, the sides, and the top at a time.
